



U6 Plant Systems Vocabulary 2025

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Xylem	tissue in plants that transports water from the roots up to the rest of the plant
Nitrogen-fixing Bacteria	bacteria that convert nitrogen in the air into forms that can be used by plants and animals
Phloem	tissue in plants that carries sugar and organic substances throughout a plant
Stimulus	a signal to which an organism responds
Vascular system	collection of specialized tissues in some plants that transports mineral nutrients and water up from the roots and brings sugars down from the leaves
Vegetative propagation	asexual reproduction in which plants produce genetically identical offshoots (clones) of themselves, which then develop into independent plants
Root hairs	tiny hair-like extensions that increase the surface area of the root allowing it to absorb more water and nutrients
Root system	all of a plant's roots, which anchor it in the soil, absorb and transport minerals and water, and store food
Shoot system	the portion of a plant above ground, consisting of stems, leaves, and flowers.
Stem	supporting structure that connects roots and leaves and carries water and nutrients between them
Leaves	the plant organs that conduct photosynthesis
Flower	the reproductive organ of a plant
Stigma	the sticky tip of a flower's female reproductive organ, the pistil/carpel
Style	stalk of a flower's pistil/carpel, with the ovary at the base and the stigma at the top
Ovary	



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a flower structure that encloses and protects ovules and seeds as they develop

Ovule	the structure within the ovary that contains the female egg cell and develops into a seed
Pollination	transfer of pollen from the male reproductive structure to the female reproductive structure
Pistil/Carpel	female reproductive organ of the flower; consists of a stigma, style, and ovary
Stamen	male reproductive organ of a flower; consists of an anther and a filament
Anther	the part of a stamen that contains the pollen
Filament	the part of a stamen that supports/holds up the anther
Auxin	a natural plant hormone that has a variety of effects including cell elongation
Stomata (stoma)	small openings/pores on the underside of a leaf through which oxygen and carbon dioxide can move
Guard cell	pairs of cells that surround stomata and control their opening and closing
Transpiration	evaporation of water from the leaves of a plant through the stomata
Phototropism	a plant's growth response to light
Gravitropism (Geotropism)	a plant's growth response to gravity
Hydrotropism	a plant's growth response to water
Thigmotropism	a plant's growth response to touch
Mesophyll cell	a plant cell type found in leaves that contain large amounts of chloroplasts; the primary site of photosynthesis



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Pollen

the tiny granules that contain the male sex cell of seed plants